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**TRINCOMALEE CAMPUS
EASTERN UNIVERSITY, SRI LANKA**

**YEAR-I SEMISTER- II
CO1225: COMPUTER ARCHITECTURE**



Questions	04
Pages	03

Trincomalee Campus, Eastern University, Sri Lanka
Faculty of Applied Science
Department of Computer Science
Bachelor of Computer Science (2017/2018)
Year I Semester II Examination (September, 2020)
CO 1225: Computer Architecture

Answer all questions

Time: 02 Hours

Q1)

a) Convert the following hexadecimal numbers into decimal, binary and octal numbers.

i) BC79DE,

ii) 16CDA.

[2*12=24 Marks]

b) Write the following decimal numbers in eight bits two's complement, do the addition/subtraction, and convert your answer back to decimal.

i) -85+11,

i) -72 + 15,

ii) -100-17.

[3*12=36 Marks]

c) Represent the following numbers in Binary Coded Decimal code.

i) 357,

ii) 8579.

[2*6=12 Marks]

d) What are the advantages of using Unicode over ASCII?

[12 Marks]

e) Convert the following binary numbers into its decimal equivalent.

i) 0.1011_2 ,

ii) 1101.0111_2 .

[2*8=16 Marks]

Q2)

- a) A logic circuit has four binary inputs D, C, B, A that represent the sixteen values 0 (0, 0, 0, 0) to 15 (1, 1, 1, 1). The output F is true if the decimal input on D, C, B, A is divisible by any of the number 3, 8, 13, or 14. Note that 0 is not divisible by any number. Draw a truth table for this system with inputs D, C, B, A and output F. [40 Marks]
- b) From the truth table, write down an expression for the output F (unsimplified). [20 Marks]
- c) Simplify the above equation obtained from part (b). [20 Marks]
- d) Design a circuit using logic gates to implement the simplified expression for F. [20 Marks]

Q3)

- a) How are computer manufacturers attempting to increase the performance of microprocessors? [15 Marks]
- b) What is a cache memory and briefly explain how it improves performance? [13 Marks]
- c) The main memory of a computer has an access time of 50 ns. The cache memory has an access time of 5 ns. The hit ratio for the memory system is 0.9 (i.e., 90% of memory accesses are to the cache). What is the speedup ratio of this system? [15 Marks]
- d) Explain the function of the program counter. [12 Marks]
- e) The operation of the simple Von Neumann computer with its fetch/execute cycle has been enhanced by several techniques to improve its performance. Write short notes on the following three mechanisms that are found in most computers. In each of the cases, explain the function using appropriate techniques.
- i) Virtual memory,
 - ii) Direct memory access (DMA),
 - iii) The interrupt.
- [15*3=45 Marks]

Q4)

a) Briefly describe the followings.

- i) ROM,
- ii) Instruction set architecture,
- iii) Pipelining,
- iv) Parallel processing,
- v) L1-Cache.

[5*08=40 Marks]

b) Briefly explain the instruction execution cycle with the suitable diagram. **[15 Marks]**

c) Draw the diagram for Von-Neumann computer architecture. **[15 Marks]**

d) What is the stored-program concept and why is it important? **[15 Marks]**

e) Compare and contrast the "Complex Instruction Set Computers (CISC)" versus "Reduced Instruction Set Computer (RISC)". **[15 Marks]**

Department



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Trincomalee Campus, Eastern University, Sri Lanka
Faculty of Applied Science
Department of Computer Science
Bachelor of Computer Science (2016/2017)
Year I Semester II Examination (February 2019)
CO1225 - Computer Architecture

Answer All Questions

Time: 02 Hours

- Q1. 1) Digital electronics only understand two states, ON and OFF. This is why digital electronics use the binary number system.
- a. Convert the following hexadecimal numbers to binary numbers:
 - i. FD7316 ;
 - ii. 75.5316 .(2x2=4 marks)
 - b. Convert 34.625 base 10 to binary, base 3, octal, and hexadecimal. Any fractions that do not terminate should be truncated to 4 digits in the fractional part.
(2x4=8 marks)
- 2) Signed numbers are required to encode negative numbers in binary number system.
- a. What are the three common ways of representing signed numbers? (2 marks)
 - b. Let the decimal numbers A=54, B= -77, give their 8-bit 2's complement representation. Compute A-B in 2's complement, is it overflow? Why or why not? (4 marks)
 - c. Give the 16-bit 2's complement representation of A, and B. Compute A-B. Is it overflow? (2 marks)
- 3) Write down whether the following statements are True or False.
- a. Registers are inside the main memory on computer mother board
 - b. Von Neumann architecture has data and instructions in the same memory space.
 - c. Dynamic RAM is faster than Static RAM in access time.
 - d. In modern computers, CPU and Memory are connected by BUS
 - e. The density of data on the hard disk does not affect data access speed.
- (1x5=5 marks)

[Total 25 Marks]

Q2. 1) For the following truth table, where x is don't care state

A	B	C	D	F
0	0	0	0	1
0	0	0	1	X
0	0	1	0	1
0	0	1	1	0
0	1	0	0	X
0	1	0	1	X
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	X
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

- a. Draw the Karnaugh map for F (4 marks)
 - b. Identify all the sub-cubes (max size, min number) on another copy of your K-map. (4 marks)
 - c. Use the sub-cubes identified in part b. to produce a minimum 'sum of products' expression for F. (3 marks)
 - d. Draw a logic the logic circuit for F (4 marks)
- 2) A full adder receives three bits (a, b, cin) as input and output the sum bit and carry bit (s, cout).
- a. Write the truth table of the full adder and the corresponding canonical product-of maxterms expressions. (5 marks)
 - b. Using basic gates draw a logic circuit for the full adder with. (5 marks)

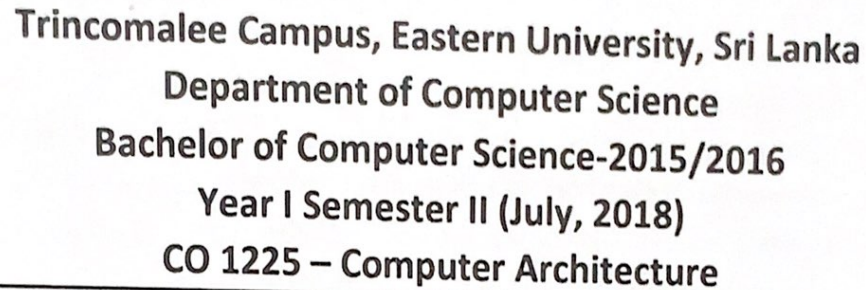
[Total 25 Marks]

- Q3. 1) A computer science graduate should have clear understanding about computer architecture and organization.
- What are the major structural components of CPU? Briefly describe.
 - Describe the functional view of a computer with the aid of a diagram.
 - Discuss the Memory Hierarchy of a computer.
 - How the computer architecture differs from Computer organization? Briefly explain.
- (3x4=12 marks)
- 2) Briefly describe the of Flynn's taxonomy for classifying computer architectures. (3 marks)
- 3) Briefly describe the following terms:
- MIPS;
 - MFLOPS;
 - Word length.
- (1x3=3 marks)
- 4) What is meant by instruction set architecture of a computer? (3 marks)
- 5) List three main addressing modes that are used on various platforms and architectures and describe them briefly. (4 marks)

[Total 25 Marks]

- Q4. 1) Pipelining is a technique used in processors.
- Describe the purpose of pipelining using suitable examples?
 - State three different types of pipeline hazards. Briefly explain any two of them.
 - Distinguish between RISC and CISC
 - Compare the Superscalar Architecture with simple pipelined architecture.
- (3x4=12 marks)
- 2) Parallel processing is a method to improve computer system performance by executing two or more instructions simultaneously.
- What are the goals of parallel processing?
 - What is Amdahl's Law?
 - Compute, if 90% of a calculation can be parallelized (i.e. 10% is sequential) then the maximum speedup, which can be achieved on 5 processors.
- (3x3=9 marks)
- 3) Briefly explain the followings:
- Shared memory;
 - Distributed Shared memory.
- (2x2=4 marks)

[Total 25 Marks]



Time Allowed: 02 Hours

Part-I

- 1) Floating point representation is used to store the,
 - a) boolean values
 - b) integers
 - c) real integers
 - d) whole numbers
- 2) In computers, generally the subtraction carried out by,
 - a) 1's complement
 - b) 2's complement
 - c) 9's complement
 - d) 10's complement
- 3) The amount of time required to read a block of data from a disk into memory is composed of seek time, rotational latency and transfer time. Rotational latency refers to
 - a) the time it takes for the platter to make a full rotation
 - b) the time it takes for the platter to rotate the correct sector under the head
 - c) the time it takes for the read-write head to move into position over the appropriate track
 - d) none of the above
- 4) The small extremely fast, RAM's are called as _____
 - a) accumulators
 - b) cache
 - c) heaps
 - d) stacks

- Page 2 of 6

- 14) The clock rate of the processor can be improved by _____
- a) by using overclocking method
 - b) improving the IC technology of the logic circuits
 - c) reducing the amount of processing done in one step
 - d) all of the above
- 15) If a processor clock is rated as 1250 million cycles per second, then its clock period is _____
- a) $1.25 * 10^{-10}$ sec
 - b) $1.6 * 10^{-9}$ sec
 - c) $1.9 * 10^{-10}$ sec
 - d) $8 * 10^{-10}$ sec
- 16) Computer memory which allows simultaneous read and write operations is
- a) ROM
 - b) RAM
 - c) EPROM
 - d) EEPROM
- 17) The circuit used to store one bit of data is known as,
- a) decoder
 - b) encoder
 - c) flip flop
 - d) register
- 18) The average time required to reach a storage location in memory and obtain its contents is called the
- a) access time
 - b) seek time
 - c) transfer time
 - d) turn around time
- 19) The idea of cache memory is based,
- a) on the fact
 - b) on the heuristic 90-10 rule
 - c) on the property of locality of reference
 - d) all of the above
- 20) The circuit used to store one bit of data is known as,
- a) decoder
 - b) encoder
 - c) flip flop
 - d) OR gate
- 21) An n-bit microprocessor has,
- a) n-bit address register
 - b) n-bit ALU
 - c) n-bit instruction register
 - d) n-bit program counter

- 22) Logic gates with a set of input and outputs is an arrangement of
- a) combinational circuit
 - b) design circuit
 - c) logic circuit
 - d) register
- 23) A group of bits that tell the computer to perform a specific operation is known as
- a) accumulator
 - b) instruction code
 - c) micro-operation
 - d) register
- 24) The load arithmetic and logical instructions are mostly used to designate a transfer from memory to a processor register known as
- a) accumulator
 - b) instruction register
 - c) memory address register
 - d) program counter
- 25) An instruction pipeline can be implemented by means of
- a) FIFO buffer
 - b) LIFO buffer
 - c) stack
 - d) none of the above

(25×4=100 Marks)

Part-II

Q1.

a) Compute the following arithmetic operations.

- | | | | |
|---------------------------|---------------------------|--------------------------|---------------------------|
| i) $4477_8 + 4421_8$ | ii) $111011_2 - 100011_2$ | iii) $7712_8 - 7644_8$ | iv) $ABC_{16} + FFF_{16}$ |
| v) $1001_2 \times 1001_2$ | vi) $54_8 \times 32_8$ | vii) $FE_{16} / 9B_{16}$ | viii) $1001_2 / 111_2$ |

(8x5=40 Marks)

b) Compute the followings using 8 bit 1's complement.

- | | | | |
|-------------|---------------|-----------------|----------------|
| i) $-5 - 7$ | ii) $10 - 12$ | iii) $-17 + 20$ | iv) $-17 + 13$ |
|-------------|---------------|-----------------|----------------|

(4x5=20 Marks)

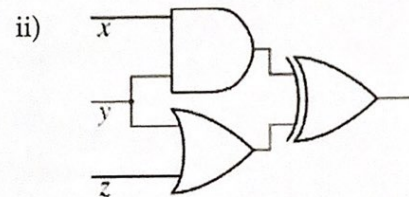
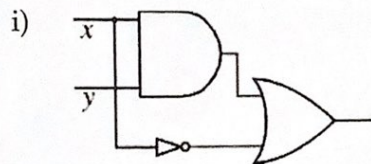
c) Compute the followings using 8 bit 2's complement.

- | | | | |
|---------------|--------------|----------------|----------------|
| i) $-25 - 15$ | ii) $13 - 7$ | iii) $-15 + 6$ | iv) $-15 - 11$ |
|---------------|--------------|----------------|----------------|

(4x5=20 Marks)

d) For each of the following circuits, write a truth table tabulating the circuit's output for each combination of inputs.

(2x5=10 Marks)



e) Consider the statement "if $(!(a < b \wedge b < c)) = 0$ ".

(2x5=10 Marks)

- i) Rewrite the if condition to an equivalent condition without using the NOT operator ("!").
- ii) Illustrates the suitable Boolean algebra law that was used for the part (i) conversion.

Q2.

- a) Explain the bus structure of a computer. (18 Marks)
- b) Compare and contrast the "Complex Instruction Set Computers (CISC)" versus "Reduced Instruction Set Computer (RISC)". (30 Marks)
- c) Discuss the performance of the instruction set architecture. (30 Marks)
- d) What is the use of registers in CPU? (05 Marks)
- e) Illustrate the operational cycle of the basic computer. (07 Marks)

- f) Differentiate the "Big endian" from "Little endian" order to store the data in computer memory. (10 Marks)

Q3.

(4x10=40 Marks)

- a) Write short notes on the following:
- Flip-Flops.
 - Multiplexer.
 - Direct Memory Access.
 - Associative Memory.
- b) Given below are the performance measures of Machine A and Machine B, which were recorded when executing a set of benchmark programmes. Assume that the clock rate of the CPU for machine A and machine B are 100 MHz and 200 MHz respectively.

Instruction category	Machine A		Machine B	
	Percentage of occurrence	No. of cycles per instruction	Percentage of occurrence	No. of cycles per instruction
ALU	38	1	30	1
Load & store	15	2	23	3
Branch	42	3	38	4
Others	5	5	9	6

Calculate the clock cycles per instruction (CPI) and the million instructions per second (MIPS) for both machines. (20 Marks)

- c) What are the reasons of "Pipe-Line" conflicts? How can they be resolved? (20 Marks)
- d) Suppose that when Program A is executed, the user CPU time is 3 seconds, the elapsed wallclock time is 4 seconds and the system performance is 10 MFLOP/sec. Assume that there are no other processes taking any significant amount of time, and the computer is either doing calculations in the CPU or doing I/O, but it can't do both at the same time. Now replace the processor with one that runs six times faster, but doesn't affect the I/O speed. What will be the user CPU time, the wallclock time and the MFLOP/sec performance? (15 Marks)
- e) Discuss about the need for parallel processing in CPU? (05 Marks)



TRINCOMALEE CAMPUS, EASTERN UNIVERSITY, SRI LANKA
DEPARTMENT OF COMPUTER SCIENCE
FIRST EXAMINATION IN COMPUTER SCIENCE-2014/2015
SECOND SEMESTER (MARCH, 2017)
CO 1225 – COMPUTER ARCHITECTURE

Answer All Questions

Time Allowed: 02 Hours

Instruction: Calculator Permitted

Part-I

Note: Select the most appropriate answer from the paper itself

1. Which representation is most efficient to perform arithmetic operations on the numbers?
a) Sign-magnitude b) 1's compliment
c) 2'S compliment d) None of the above
2. When we perform subtraction on -7 and 1 the answer in 2's compliment form is _____.
a) 1010 b) 1110 c) 0110 d) 1000
3. When 1101 is used to divide 100010010 the remainder is _____.
a) 101 b) 11 c) 0 d) 1
4. To reduce the memory access time we generally make use of _____.
a) Heaps b) Higher capacity RAM's c) SDRAM's d) Cache's
5. The smallest entity of memory is called as _____.
a) Cell b) Block c) Instance d) Unit
6. _____ is generally used to increase the apparent size of physical memory .
a) Secondary memory b) Virtual memory
c) Hard-disk d) Disks

7. An 24 bit address generates an address space of _____ locations .
a) 1024 b) 4096 c) 2^{48} d) 16,777,216
8. _____ method is used to map logical addresses of variable length onto physical memory.
a) Paging b) Overlays
c) Segmentation d) Paging with segmentation
9. During transfer of data between the processor and memory we use _____.
a) Cache b) TLB c) Buffers d) Registers
10. Physical memory is divided into sets of finite size called as _____.
a) Frames b) Pages c) Blocks d) Vectors
11. The virtual memory basically stores the next segment of data to be executed on the _____.
a) Secondary storage b) Disks c) RAM d) ROM
12. ISP stands for,
a) Instruction Set Processor b) Information Standard Processing
c) Interchange Standard Protocol d) Interrupt Service Procedure
13. During the execution of the instructions, a copy of the instructions is placed in the _____.
a) Register b) RAM c) System heap d) Cache
14. The instructions like MOV or ADD are called as _____.
a) OP-Code b) Operators c) Commands d) None of the above
15. When Performing a looping operation, the instruction gets stored in the _____.
a) Registers b) Cache c) System Heap d) System stack
16. The last statement of the source program should be _____.
a) Stop b) Return c) OP d) End
17. The order in which the return addresses are generated and used is _____.
a) LIFO b) FIFO c) Random d) Highest priority

- (25x4=100 Marks)**

Part-II

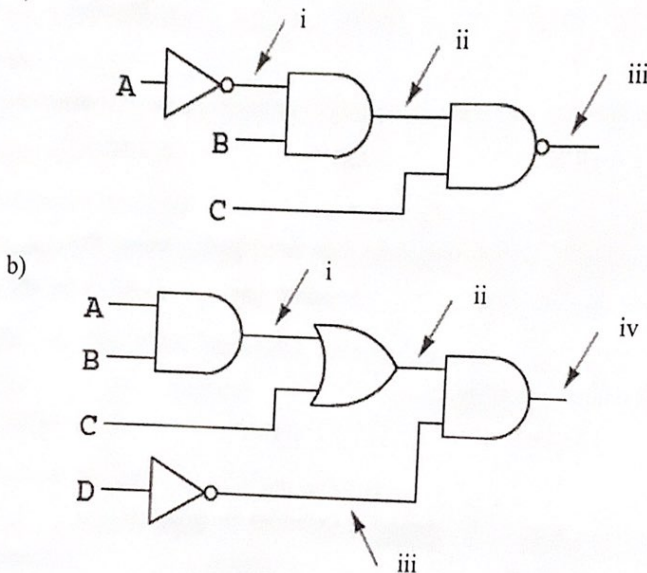
Q1.

- i. Compute the following arithmetic operations.

a. $3477_8 + 7744_8$	b. $1111011_2 - 110001_2$	c. $7722_8 - 7643_8$	d. $9BC_{16} + 77F_{16}$
e. $1001_2 \times 10011_2$	f. $754_8 \times 320_8$	g. $AC_{16} \times DF_{16}$	h. $FFEE_{16} / AB_{16}$
i. $100011_2 / 1111_2$	j. $77566_8 / 765_8$		
- ii. Compute the followings using 8 bit 1's complement.

a. $-7 - 9$	b. $-18 + 20$	c. $-17 + 15$	d. $-13 - 11$
-------------	---------------	---------------	---------------
- iii. Compute the followings using 8 bit 2's complement.

a. $-21 - 15$	b. $-18 + 5$	c. $-12 - 13$	d. $13 - 19$
---------------	--------------	---------------	--------------
- iv. Show that the dual of XOR is also its complement. (6 Marks)
- v. Convert the following logic gate circuits into a Boolean expression. Write down the suitable Boolean sub-expressions for the arrow indicators in the circuit.



(7x2=14 Marks)

Q2.

- i. Briefly discuss the different mappings employed in cache memory. (45 Marks)
- ii. Clearly explain the different cache writing policies. (25 Marks)
- iii. Explain the different techniques used for interfacing I/O units with the processor. (30 Marks)

Q3.

- i. Briefly explain the term "Virtual Memory". **(10 Marks)**
- ii. Why page-table is required in a virtual memory system? Explain different ways of organizing a page table. **(25 Marks)**
- iii. If a virtual-to-physical address translation takes ~100 clock cycles, what is the most likely reason for this particular latency? **(15 Marks)**
- iv. What do you mean by "Pipe-Line"? **(10 Marks)**
- v. What are the reasons of Pipe-Line conflicts in a Pipe Lined processor? How are they resolved? **(20 Marks)**
- vi. Formulate a four segment instruction pipeline for a computer. Specify the operation to be performed in each segment. **(20 Marks)**